# BBST Trees

Applications of AVL trees

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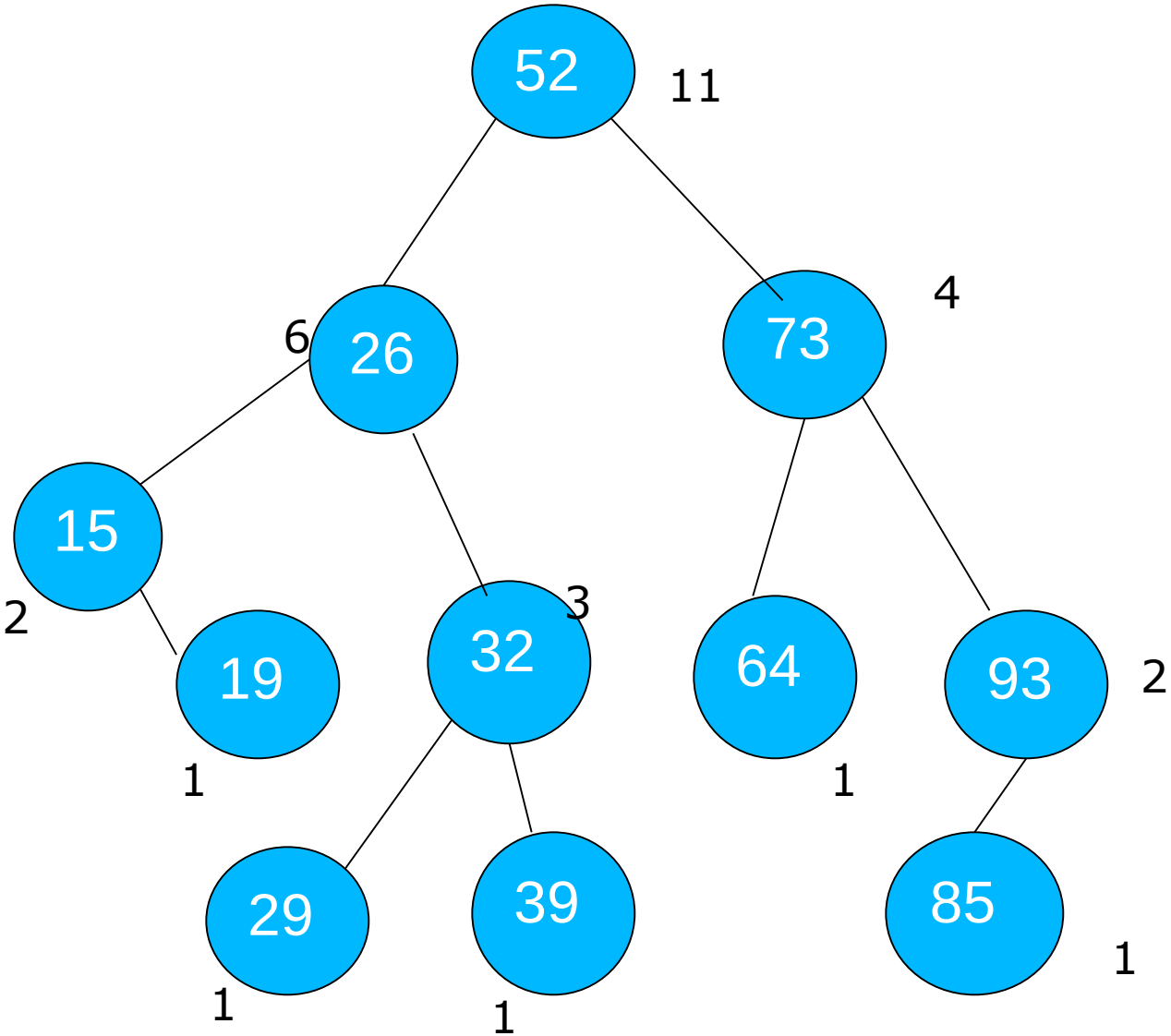
# Rank (X)

Implement a Data Structure which supports the following operations efficiently.

- Search
- Insert
- Delete

- Rank(X) – 1+ number of numbers in the collection that are >X.

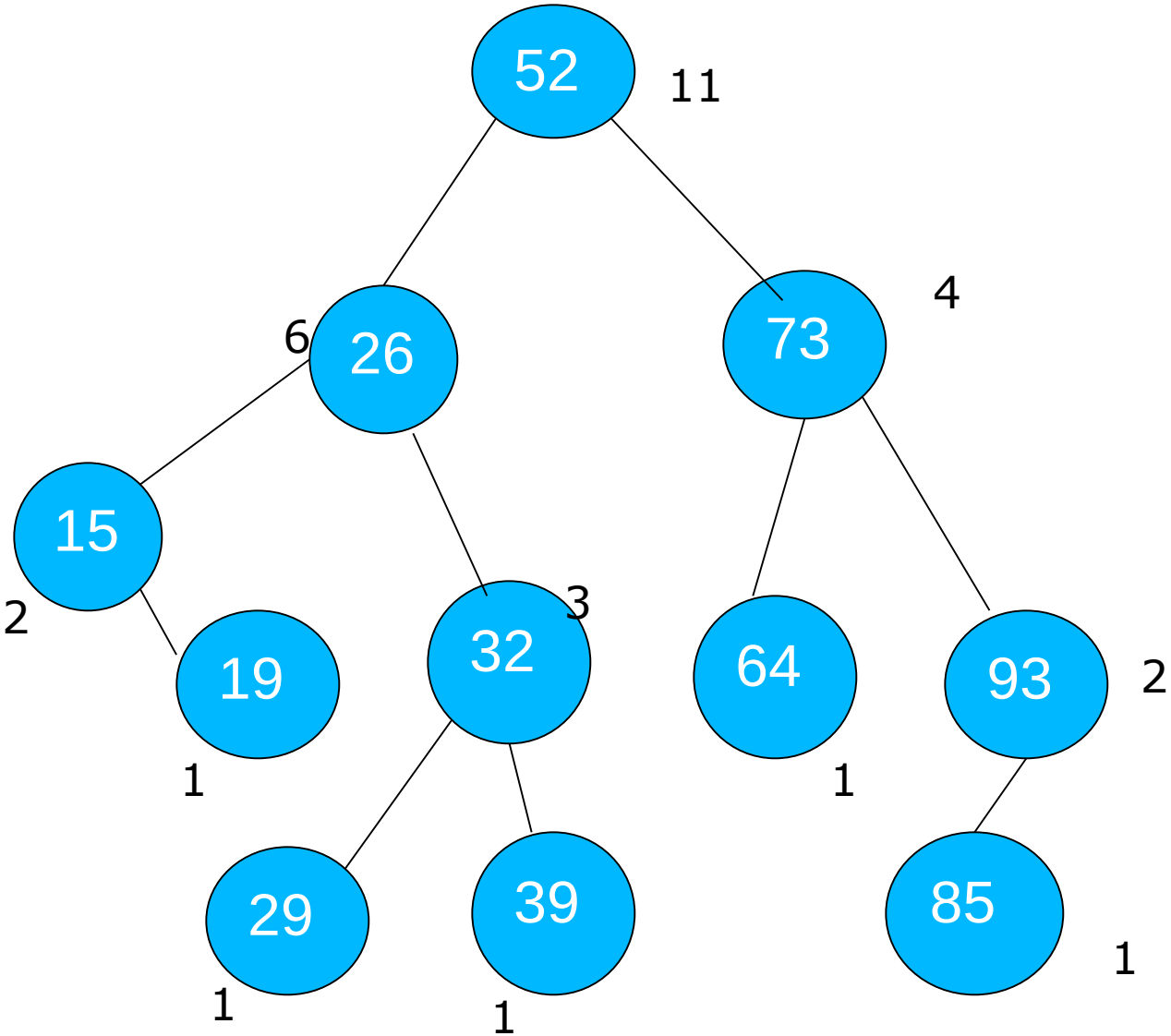
# Node count



# Find Rank (rank)

Given rank , find an number X, such Rank(X)= r

# Node count



# Other operations

- PrefixSum(X) = find the sum of all the numbers smaller than X.

# Other operations

- PrefixSum(X) = find the sum of all the numbers smaller than X.
- RangeSum(X,Y) = sum of numbers between X and Y.
- Average(X,Y) = Average of numbers between X and Y.
- Morethanaverage(X,Y) = the no of numbers between X and Y bigger than Average(X,Y).

# Range Query

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 4 | 8 | 2 | 5 | 1 | 6 |

RangeMaxQuery(A,0,4)=1;

RangeMinQuery(A,0,3)=2;

Update(A,i ,X) = Set A[i]=X;

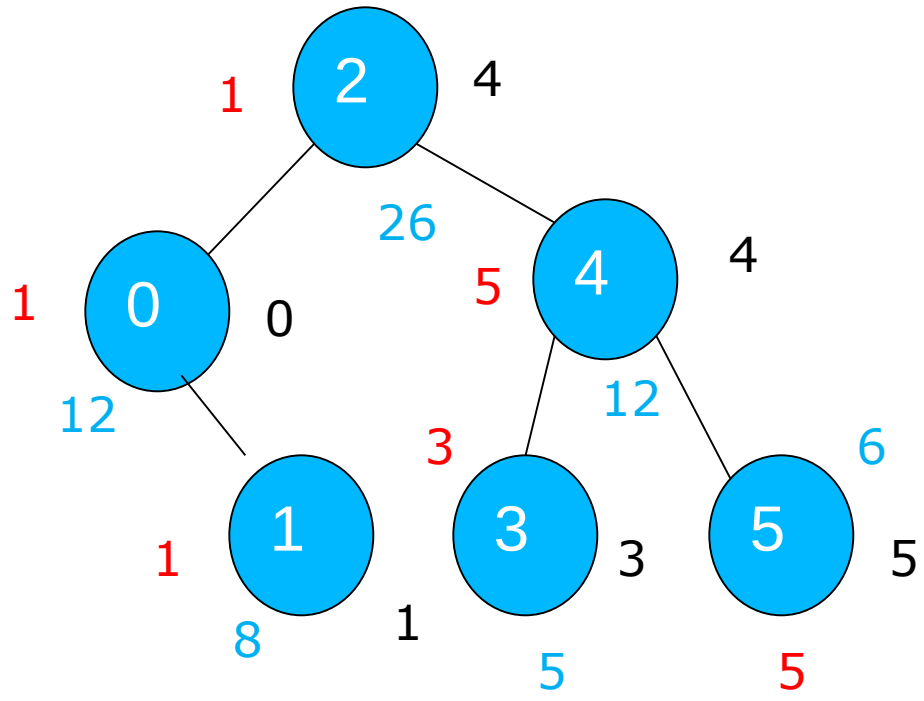
# Range Query

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 4 | 8 | 2 | 5 | 1 | 6 |

min

max

sum



# BBST Structure

Index are the keys in the BST.

The BBST can be built in linear time .

# BBST in O(log n)

- Update()
- PreSum()
- RangeSum()
- RangeMinQuery()
- RangeMaxQuery()
- RangeMaxGap() –maximum difference among all the numbers in the given range

# Dynamic Data Set

- Add()
- Delete()
- Search()
- Mingap() minimum difference among the numbers
- Maxgap() maximum difference among the numbers – is the difference between max and min.

# Dynamic Data Set

2, 15, 32, 35, 55

- Mingap()=3
- Maxgap()=53

# Dynamic Data Set

2, 15, 32, 35, 55

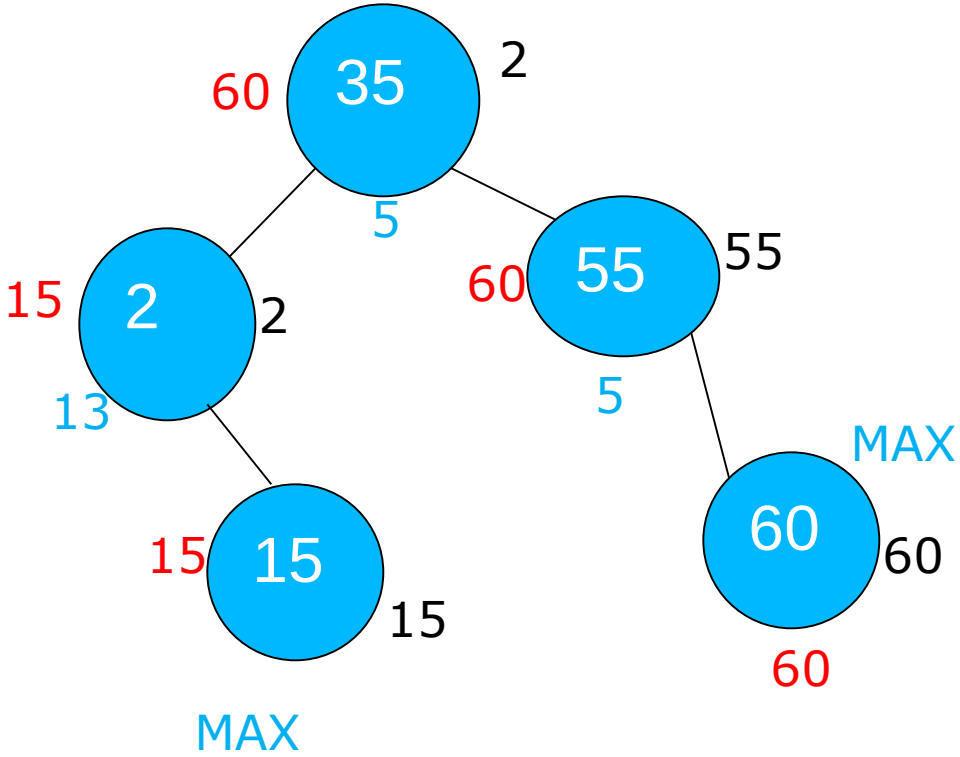
- Mingap()=3
- Maxgap()=53
- Delete(32) and Add(60)
- 2, 15, 35, 55,60
- Mingap()=5
- Maxgap()=58

# Range Query

min

max

mingap



# Max value at a node

Node->max=Max

- Node->value
- Node->left->max
- Node->right->max

# Max value at a node

Node->min=Min

Node->value

Node->left->min

Node->right->min

# Max value at a node

node->
mingap=Min

node->value-node->left->max

node->right->min-node->value

node->left->mingap

node->right->mingap

- Add()
- Delete()
- Search()
- RangeMaxGap()
- RangeMinGap()

- Maxgap and Mingap in the whole set can be found on O(1)